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Visual Inspection System for a Medical Device

Abstract

A visual inspection system (**600**) for a syringe (**502**) that includes an image capture device (**104**), a structure configured to hold a tube of the syringe (**502**) for inspection, and at least one processor (**102**) that may cause an image of the tube to be captured via the image capture device when the tube is positioned in the structure, wherein the image includes an image of a flange (**510**) of the tube superimposed on a background (**404**), wherein the background includes white and black. The processor may determine a diameter of the flange of the tube and provide data associated with the diameter of the flange of the tube.

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Background/Summary

BACKGROUND

[0001] Machine vision (MV) may refer to technology used to provide imaging-based automatic inspection and/or analysis for industrial applications, such as automatic inspection, process control, and/or robot guidance. Machine vision may include functions in industrial automation environments but may also include functions in other environments, such as security and vehicle guidance. An overall machine vision process may include planning the details of requirements for a project and then creating a solution. During run-time, a machine vision process may start with an imaging process, followed by automated analysis of an image, and then extraction of required information from the automated analysis.

[0002] However, in certain situations, an imaging process may fail to provide an image which is viable for extracting information from an image. For example, an image of a medical device, such as a syringe or other container, may include a background that does not allow for accurate detection of particular aspects of the medical device from the image.

SUMMARY

[0003] Accordingly, provided are improved systems, devices, products, apparatus, and/or methods for inspecting a flange and/or internal diameter of a medical device.

[0004] Further non-limiting embodiments or aspects are set forth in the following numbered clauses: [0005] Clause 1: A method for inspecting aspects of a medical device comprising: causing, with at least one processor, an image of a tube to be captured via an image capture device, wherein the image comprises an image of a flange of the tube superimposed on a background, wherein the background comprises a first color and a second color; determining, with at least one processor, a measurement of an attribute of the flange of the tube; and providing, with at least one processor, data associated with the measurement of an attribute of the flange of the tube. [0006] Clause 2: The method of clause 1, further comprising: performing an edge detection procedure on the image of the flange of the tube to provide a defined edge of the flange of the tube; wherein determining the measurement of the attribute of the flange of the tube comprises: determining the measurement of the attribute of the flange of the tube based on the defined edge of the flange of the tube. [0007] Clause 3: The method of clause 1 or 2, further comprising: positioning the tube within a structure, wherein a surface of the structure comprises the first color and the second color. [0008] Clause 4: The method of any of clauses 1-3, wherein the structure comprises a first component and a second component, and wherein the structure is configured to hold the tube between the first component and the second component. [0009] Clause 5: The method of any of clauses 1-4, further comprising: causing light to illuminate the tube via at least one light source prior to causing the image of the tube to be captured via an image capture device, wherein the at least one light source comprises a spot light and a ring light. [0010] Clause 6: The method of any of clauses 1-5, wherein the first color of the background is white and wherein the second color of the background is black. [0011] Clause 7: The method of any of clauses 1-6, wherein the tube is a syringe. [0012] Clause 8: A system for inspecting aspects of a medical device, comprising: a structure configured to hold a tube for inspection; and at least one processor programmed or configured to: cause an image of a tube to be captured via an image capture device when the tube is positioned in the structure, wherein the image comprises an image of a flange of the tube superimposed on a background, wherein the background comprises a first color and a second color; determine a measurement of an attribute of the flange of the tube; and provide data associated with the measurement of an attribute of the flange of the tube. [0013] Clause 9: The system of clause 8, further comprising: at least one light source, wherein the at least one light source is configured to illuminate the flange of the tube. [0014] Clause 10: The system of clause 8 or 9, wherein the at least one processor is further

programmed or configured to: perform an edge detection procedure on the image of the flange of the tube to provide a defined edge of the flange of the tube; wherein, when determining the measurement of the attribute of the flange of the tube, the at least one processor is programmed or configured to: determine the measurement of the attribute of the flange of the tube based on the defined edge of the flange of the tube. [0015] Clause 11: The system of any of clauses 8-10, further comprising: a tube positioned within the structure, wherein a surface of the structure comprises the first color and the second color. [0016] Clause 12: The system of any of clauses 8-11, wherein the structure comprises a first component and a second component, and wherein the structure is configured to hold the tube between the first component and the second component. [0017] Clause 13: The system of any of clauses 8-12, wherein the at least one processor is further programmed or configured to: cause light to illuminate the tube via the at least one light source prior to causing the image of the tube to be captured via an image capture device, wherein the at least one light source comprises a spot light and a ring light. [0018] Clause 14: The system of any of clauses 8-13, wherein the first color of the background is white and wherein the second color of the background is black. [0019] Clause 15: The system of any of clauses 8-14, wherein the tube is a syringe. [0020] Clause 16: A system for inspecting aspects of a medical device, comprising: a structure configured to hold a medical device for inspection; an image capture device; and at least one processor programmed or configured to: cause an image of a tube to be captured via the image capture device when the tube is positioned in the structure, wherein the image comprises an image of a flange of the tube superimposed on a background, wherein the background comprises a first color and a second color; determine a measurement of an attribute of the flange of the tube; and provide data associated with the measurement of an attribute of the flange of the tube. [0021] Clause 17: The system of clause 16, wherein the image capture device is positioned to capture the image of the tube when the flange of the tube is illuminated via at least one light source, wherein the at least one light source comprises a spot light and a ring light. [0022] Clause 18: The system of clause 16 or 17, further comprising: a tube positioned within a structure, wherein a surface of the structure comprises the first color and the second color. [0023] Clause 19: The system of any of clauses 16-18, wherein the structure comprises a first component and a second component, and wherein the structure is configured to hold the tube between the first component and the second component. [0024] Clause 20: The system of any of clauses 16-19, wherein the first color of the background is white and wherein the second color of the background is black.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Additional advantages and details are explained in greater detail below with reference to the exemplary embodiments that are illustrated in the accompanying schematic figures, in which:

[0026] FIG. 1 is a diagram of a non-limiting embodiment or aspect of an environment in which methods and/or systems, described herein, may be implemented according to the principles of the presently disclosed subject matter;

[0027] FIG. 2 is a diagram of a non-limiting embodiment of components of one or more devices and/or one or more systems of FIG. 1;

[0028] FIG. 3 is a flowchart of a non-limiting embodiment of a process for visually inspecting a medical device;

[0029] FIG. 4A is a perspective view of a non-limiting embodiment of a structure for an inspection system;

[0030] FIG. 4B is a front view of a non-limiting embodiment of a structure for an inspection system;

[0031] FIG. 5 is a diagram of a non-limiting embodiment of a medical device used in an inspection

system;

[0032] FIG. 6A is a diagram of a non-limiting embodiment of an implementation of an inspection system;

[0033] FIG. 6B is a diagram of a non-limiting embodiment of another implementation of an inspection system; and

[0034] FIGS. 7A and 7B are views of a non-limiting embodiment of captured images of an inspection system.

DETAILED DESCRIPTION

[0035] It is to be understood that the present disclosure may assume various alternative variations and step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary and non-limiting embodiments or aspects. Hence, specific dimensions and other physical characteristics related to the embodiments or aspects disclosed herein are not to be considered as limiting.

[0036] For purposes of the description hereinafter, the terms “end,” “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “lateral,” “longitudinal,” and derivatives thereof shall relate to embodiments or aspects as they are oriented in the drawing figures. However, it is to be understood that embodiments or aspects may assume various alternative variations and step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply non-limiting exemplary embodiments or aspects. Hence, specific dimensions and other physical characteristics related to the embodiments or aspects of the embodiments or aspects disclosed herein are not to be considered as limiting unless otherwise indicated.

[0037] No aspect, component, element, structure, act, step, function, instruction, and/or the like used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more” and “at least one.” As used in the specification and the claims, the singular form of “a,” “an,” and “the” include plural referents, such as unless the context clearly dictates otherwise. Additionally, Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items (e.g., related items, unrelated items, a combination of related and unrelated items, etc.) and may be used interchangeably with “one or more” or “at least one.” Where only one item is intended, the term “one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based at least partially on” unless explicitly stated otherwise. Further, the phrase “based on” is intended to mean “based at least partially on” unless explicitly stated otherwise. Further, the phrase “based on” may mean “in response to” and be indicative of a condition for automatically triggering a specified operation of an electronic device (e.g., a controller, a processor, a computing device, etc.) as appropriately referred to herein.

[0038] As used herein, the terms “communication” and “communicate” may refer to the reception, receipt, transmission, transfer, provision, and/or the like of information (e.g., data, signals, messages, instructions, commands, and/or the like). For one unit (e.g., a device, a system, a component of a device or system, combinations thereof, and/or the like) to be in communication with another unit means that the one unit is able to directly or indirectly receive information from and/or send (e.g., transmit) information to the other unit. This may refer to a direct or indirect connection that is wired and/or wireless in nature. Additionally, two units may be in communication with each other even though the information transmitted may be modified, processed, relayed, and/or routed between the first and second unit. For example, a first unit may be in communication with a second unit even though the first unit passively receives information and does not actively transmit information to the second unit. As another example, a first unit may

be in communication with a second unit if at least one intermediary unit (e.g., a third unit located between the first unit and the second unit) processes information received from the first unit and transmits the processed information to the second unit. In some non-limiting embodiments or aspects, a message may refer to a network packet (e.g., a data packet and/or the like) that includes data.

[0039] Some non-limiting embodiments or aspects may be described herein in connection with thresholds. As used herein, satisfying a threshold may refer to a value being greater than the threshold, more than the threshold, higher than the threshold, greater than or equal to the threshold, less than the threshold, fewer than the threshold, lower than the threshold, less than or equal to the threshold, equal to the threshold, etc.

[0040] Provided are improved devices, systems, methods and computer program products for inspecting a flange and/or internal diameter of a medical device. Embodiments of the present disclosure may include an inspection system that includes a structure, an image capture device, and at least one processor in communication with the image capture device, wherein the at least one processor is programmed or configured to cause an image of a tube to be captured via the image capture device when a tube is positioned in the structure, wherein the image comprises an image of a flange of the tube superimposed on a background, wherein the background comprises a first color and a second color, determine a measurement of an attribute of the flange of the tube, and provide data associated with the measurement of an attribute of the flange of the tube. In some non-limiting embodiments or aspects, a surface of the structure may include the first color and the second color.

[0041] In some non-limiting embodiments, the at least one processor is further programmed or configured to perform an edge detection procedure on the image of the flange of the tube to provide a defined edge of the flange of the tube, wherein determining the measurement of the attribute of the flange of the tube includes determining the measurement of the attribute of the flange of the tube based on the defined edge of the flange of the tube. In some non-limiting embodiments or aspects, the at least one processor is further programmed or configured to cause light to illuminate the tube via at least one light source prior to causing the image of the tube to be captured via an image capture device, wherein the at least one light source comprises a spot light and a ring light.

[0042] In this way, embodiments of the present disclosure allow for an inspection system that may accurately detect particular aspects of a medical device, such as a tube, a syringe, and/or the like, from an image. For example, the inspection system may be able to determine the presence of defects and/or more accurately detect an internal diameter of the medical device to ensure proper inspection of the medical device.

[0043] Referring now to FIG. 1, FIG. 1 is a diagram of a non-limiting embodiment or aspect of an environment **100** in which methods and/or systems, as described herein, may be implemented. As shown in FIG. 1, environment **100** may include inspection system **102**, image capture device **104**, and light source **106**. Inspection system **102**, image capture device **104**, and light source **106** may interconnect (e.g., establish a connection to communicate, and/or the like) via wired connections, wireless connections, or a combination of wired and wireless connections. In some non-limiting embodiments, inspection system **102** may include one or more of image capture device **104**, and/or light source **106**.

[0044] In some non-limiting embodiments, inspection system **102** may include one or more devices capable of receiving information from and/or communicating information to image capture device **104**, and/or light source **106**. For example, inspection system **102** may include a computing device, such as a processor, a server, a group of servers, and/or other like devices. Inspection system **102** may be capable of sending signals to image capture device **104** and/or light source **106** and inspection system **102** may be capable of receiving data and/or signals from image capture device **104** and/or light source **106**.

[0045] In some non-limiting embodiments, image capture device **104** may include a camera. Image capture device **104** may be capable of capturing static pictures and/or dynamic pictures (e.g.,

video). In some non-limiting embodiments, image capture device **104** may be capable of receiving information from and/or communicating information to inspection system **102**. For example, image capture device **104** may be capable of receiving signals from inspection system **102** and/or may be capable of sending signals and/or data to inspection system **102**.

[0046] In some non-limiting embodiments, light source **106** may include a spot light and/or a ring light. Light source **106** may be adjusted such that light source **106** may illuminate one or more parts of inspection system **102**. In some non-limiting embodiments, light source **106** may be capable of receiving information from and/or communicating information to inspection system **102**. For example, light source **106** may be capable of receiving signals from inspection system **102** and/or may be capable of sending signals and/or data to inspection system **102**. In some non-limiting embodiments, light source **106** may include one or more light sources (e.g., one or more spot lights and/or one or more ring lights).

[0047] Referring now to FIG. 2, FIG. 2 is a diagram of example components of device **200**. Device **200** may correspond to inspection system **102**, image capture device **104**, light source **106**, and/or one or more components of inspection system **102**. In some non-limiting embodiments, inspection system **102**, image capture device **104**, and/or light source **106** may include at least one device **200** and/or at least one component of device **200**. As shown in FIG. 2, device **200** may include bus **202**, processor **204**, memory **206**, storage component **208**, input component **210**, output component **212**, and communication interface **214**.

[0048] Bus **202** may include a component that permits communication among the components of device **200**. In some non-limiting embodiments, processor **204** may be implemented in hardware, software, or a combination of hardware and software. For example, processor **204** may include a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), etc.), a microprocessor, a digital signal processor (DSP), and/or any processing component (e.g., a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), etc.) that can be programmed to perform a function. Memory **206** may include random access memory (RAM), read-only memory (ROM), and/or another type of dynamic or static storage device (e.g., flash memory, magnetic memory, optical memory, etc.) that stores information and/or instructions for use by processor **204**.

[0049] Storage component **208** may store information and/or software related to the operation and use of device **200**. For example, storage component **208** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, a solid state disk, etc.), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, and/or another type of computer-readable medium, along with a corresponding drive.

[0050] Input component **210** may include a component that permits device **200** to receive information, such as via user input (e.g., a touchscreen display, a keyboard, a keypad, a mouse, a button, a switch, a microphone, a camera, etc.). Additionally or alternatively, input component **210** may include a sensor for sensing information (e.g., a global positioning system (GPS) component, an accelerometer, a gyroscope, an actuator, etc.). Output component **212** may include a component that provides output information from device **200** (e.g., a display, a speaker, one or more light-emitting diodes (LEDs), etc.).

[0051] Communication interface **214** may include a transceiver-like component (e.g., a transceiver, a separate receiver and transmitter, etc.) that enables device **200** to communicate with other devices, such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. Communication interface **214** may permit device **200** to receive information from another device and/or provide information to another device. For example, communication interface **214** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a Wi-Fi® interface, a Bluetooth® interface, a Zigbee® interface, a cellular network interface, and/or the like.

[0052] Device **200** may perform one or more processes described herein. Device **200** may perform

these processes based on processor **204** executing software instructions stored by a computer-readable medium, such as memory **206** and/or storage component **208**. A computer-readable medium (e.g., a non-transitory computer-readable medium) is defined herein as a non-transitory memory device. A non-transitory memory device includes memory space located inside of a single physical storage device or memory space spread across multiple physical storage devices.

[0053] Software instructions may be read into memory **206** and/or storage component **208** from another computer-readable medium or from another device via communication interface **214**. When executed, software instructions stored in memory **206** and/or storage component **208** may cause processor **204** to perform one or more processes described herein. Additionally or alternatively, hardwired circuitry may be used in place of or in combination with software instructions to perform one or more processes described herein. Thus, embodiments or aspects described herein are not limited to any specific combination of hardware circuitry and software.

[0054] Memory **206** and/or storage component **208** may include data storage or one or more data structures (e.g., a database and/or the like). Device **200** may be capable of receiving information from, storing information in, communicating information to, or searching information stored in the data storage or one or more data structures in memory **206** and/or storage component **208**. For example, the information may include input data, output data, image data, measurement data, or any combination thereof.

[0055] The number and arrangement of components shown in FIG. **2** are provided as an example. In some non-limiting embodiments, device **200** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **2**. Additionally or alternatively, a set of components (e.g., one or more components) of device **200** may perform one or more functions described as being performed by another set of components of device **200**.

[0056] Referring now to FIG. **3**, FIG. **3** is a flowchart of a non-limiting embodiment or aspect of process **300** for inspecting aspects of a medical device. In some non-limiting embodiments, one or more of the functions described with respect to process **300** may be performed (e.g., completely, partially, etc.) by inspection system **102**. In some non-limiting embodiments, one or more of the steps of process **300** may be performed (e.g., completely, partially, and/or the like) by another device or a group of devices separate from and/or including inspection system **102**, such as image capture device **104**, and/or light source **106**.

[0057] As shown in FIG. **3**, at step **302**, process **300** may include causing an image to be captured. For example, inspection system **102** may cause an image to be captured with image capture device **104**. In some non-limiting embodiments, the image associated with image capture device **104** may include an image of a flange of a tube (e.g., syringe) superimposed on a background. In some non-limiting embodiments, the image associated with image capture device **104** may include an axial (e.g., cross-sectional) view of the tube superimposed on a background. In some non-limiting embodiments, the background may include a first color and a second color. In some non-limiting embodiments, the first color may be white and the second color may be black.

[0058] In some non-limiting embodiments, inspection system **102** may store the image as image data that is determined (e.g., measured, detected, sensed, obtained, etc.) by inspection system **102** in the course of operation of inspection system **102**.

[0059] As shown in FIG. **3**, at step **304**, process **300** may include determining a measurement. For example, inspection system **102** may determine a measurement associated with the flange of the tube based on the image captured by image capture device **104**. In some non-limiting embodiments, inspection system **102** may determine a measurement of an attribute of the flange of the tube. For example, inspection system **102** may determine the measurement of an internal diameter of the flange of the tube. In some non-limiting embodiments, inspection system **102** may determine a measurement of an attribute of the tube. In some non-limiting embodiments, inspection system **102** may perform an inspection on the tube after the tube is positioned within a structure where the

surface of the structure includes the first color and the second color.

[0060] In some non-limiting embodiments, determining the measurement of the attribute of the flange of the tube may include determining the measurement of the attribute of the flange of the tube based on the defined edge of the flange of the tube.

[0061] In some non-limiting embodiments, inspection system **102** may determine the measurement associated with the flange of the tube by analyzing the image and/or image data captured by image capture device **104**.

[0062] As shown in FIG. 3, at step **306**, process **300** may include providing data. For example, inspection system **102** may provide data associated with the measurement of an attribute of the flange of the tube. In some non-limiting embodiments, the data associated with the measurement of an attribute may include a value of an internal diameter of the flange of the tube.

[0063] In some non-limiting embodiments, inspection system **102** may cause light to illuminate the tube via at least one light source (e.g., light source **106**) prior to causing the image of the tube to be captured via image capture device **104**. In some non-limiting embodiments, the at least one light source may include a spot light and/or a ring light. In some non-limiting embodiments, light source **106** may include one or more light sources (e.g., one or more spot lights and/or one or more ring lights).

[0064] With regard to FIGS. 4A and 4B, FIGS. 4A and 4B show a perspective view of a non-limiting embodiment of a structure and a front view of a non-limiting embodiment of a structure for inspection system **102**. Inspection system **102** may include a structure according to non-limiting embodiments of the present disclosure.

[0065] Referring now to FIG. 4A, FIG. 4A is a perspective view of a non-limiting embodiment of a structure **400** for an inspection system. As shown in FIG. 4A, structure **400** may include at least two support blocks **402**. In some non-limiting embodiments, structure **400** may be configured to hold a tube between a first support block **402** and a second support block **402**.

[0066] In some non-limiting embodiments, support block **402** may include front surface **404**, back surface **406**, first side surface **408**, second side surface **410**, top surface **412**, bottom surface **414**, bored holes **416**, cutouts **418**, fastener holes **420**, through hole **422**, first color **424**, and second color **426**. In some non-limiting embodiments, a first portion of support block **402** may include first color **424** and a second portion of support block **402** may include second color **426**.

[0067] In some non-limiting embodiments, front surface **404** may include first color **424** and second color **426**. In some non-limiting embodiments, front surface **404** may be a background in an image of a flange superimposed on the background. In some non-limiting embodiments, back surface **406** and top surface **412** may include first color **424** and second color **426**. In some non-limiting embodiments, first side surface **408**, second side surface **410**, and bottom surface may include second color **426**. In some non-limiting embodiments, front surface **404** may include a length equal to 30 millimeters (mm) and a width equal to 16.3 mm. In some non-limiting embodiments, front surface **404** may include a length up to 16 mm. In some non-limiting embodiments, back surface **406** may include a length equal to the length of front surface **404** and a width equal to the width of front surface **404**.

[0068] In some non-limiting embodiments, first side surface **408** may include a length equal to 16.3 mm and a width equal to 16 mm. In some non-limiting embodiments, first side surface **408** may include a length up to 18 mm. In some non-limiting embodiments, second side surface **410** may include a length equal to the length of first side surface **408** and a width equal to the width of first side surface **408**.

[0069] In some non-limiting embodiments, top surface **412** may include one or more bored holes **416** and cutout **418**. In some non-limiting embodiments, bored hole **416** may be configured to accept a fastener. In some non-limiting embodiments, cutout **418** may include a cylindrical cutout, a semi-cylindrical cutout, a square cutout, or another shape such that cutout **418** may be configured to accept a tube or other object positioned within the space defined by cutout **418**. In some non-

limiting embodiments, cutout **418** may include a diameter equal to 8.26 mm. In some non-limiting embodiments, cutout **418** may extend from front surface **404** to back surface **406**. In some non-limiting embodiments, cutout **418** may accept a tube when at least one first cutout **418** is merged with at least one second cutout **418** when two support blocks **402** are brought together such that top surfaces **412** of each support block **402** contact each other. In some non-limiting embodiments, cutout **418** may include through-hole **422**. In some non-limiting embodiments, cutout **418** may include first color **424**.

[0070] In some non-limiting embodiments, top surface **412** may include a length equal to 30 mm and a width equal to 16 mm. In some non-limiting embodiments, bottom surface **414** may include a length equal to the length of top surface **412** and a width equal to the width of top surface **412**.

[0071] In some non-limiting embodiments, bottom surface **414** may include one or more fastener holes **420** and through-hole **422**. In some non-limiting embodiments, fastener hole **420** may be configured to accept a fastener. In some non-limiting embodiments, through-hole **422** may extend from top surface **412** through support block **402** to bottom surface **414**.

[0072] In some non-limiting embodiments, bored hole **416** may include a first hole including a first diameter and a bore including a bore depth and a bore diameter. In some non-limiting embodiments, the first hole may include a first diameter equal to 3.4 millimeters (mm). In some non-limiting embodiments, the bore may include a bore depth equal to 3.4 mm and a bore diameter equal to 6.5 mm.

[0073] In some non-limiting embodiments, fastener hole **420** may include a countersink equal to 90 degrees and having a depth of 1 mm. In some non-limiting embodiments, fastener hole **420** may include a first hole portion including a depth equal to 7.3 mm and a diameter equal to 2.4 mm. In some non-limiting embodiments, fastener hole **420** may include a second hole portion including a depth between 3 mm and 4.48 mm. The depth of the second hole portion may be measured from the deepest point of the first hole portion. For example, fastener hole **420** may have a total depth of 10.3 mm to 11.8 mm. In some non-limiting embodiments, the second hole portion may be threaded in order to accept a threaded fastener.

[0074] In some non-limiting embodiments, through hole **422** may include a total depth equal to the length of support block **402** (e.g., the length of front surface **404**, back surface **406**, first side surface **408**, and/or second side surface **410**). In some non-limiting embodiments, through hole **422** may include a diameter of 3.5 mm.

[0075] In some non-limiting embodiments, bored holes **416** may be located along the center line of the width of top surface **412** and bottom surface **414** of support block **402**. Alternatively, bored holes **416** may be located offset from the center line of the width of top surface **412** and bottom surface **414**. For example, bored holes **416** may be located 12 mm away from the edge of front surface **404** and 4 mm away from the edge of back surface **406** where the length of top surface **412** and bottom surface **414** is 16 mm (e.g., the center line of the width is located at 8 mm from the edge of top surface **412** and bottom surface **414**). In some non-limiting embodiments, a first support block **402** may have bored holes **416** offset from the center line of the width of top surface **412** and bottom surface **414** and a second support block **402** may have bored holes **416** located along the center line of the width of top surface **412** and bottom surface **414**.

[0076] In some non-limiting embodiments, bottom surface **414** may include one fastener hole **420**. In some non-limiting embodiments, fastener holes **420** may be located along the center line of the width of top surface **412** and bottom surface **414** of support block **402**. Alternatively, fastener holes **420** may be located offset from the center line of the width of top surface **412** and bottom surface **414**. For example, fastener holes **420** may be located 12 mm away from the edge of front surface **404** and 4 mm away from the edge of back surface **406** where the length of top surface **412** and bottom surface **414** is 16 mm (e.g., the center line of the width is located at 8 mm from the edge of top surface **412** and bottom surface **414**). In some non-limiting embodiments, further fastener holes may be located 12 mm away from the edge of bottom surface **414** and 4 mm away from the edge of

top surface **412**. In some non-limiting embodiments, a first support block **402** may have fastener holes **420** offset from the center line of the width of top surface **412** and bottom surface **414** and a second support block **402** may have fastener holes **420** located along the center line of the width of top surface **412** and bottom surface **414**.

[0077] In some non-limiting embodiments, through hole **422** may be located along the center line of the width of top surface **412** and bottom surface **414** of support block **402**. Alternatively, through-hole **422** may be located offset from the center line of the width of top surface **412** and bottom surface **414**. For example, through-hole **422** may be located 12 mm away from the edge of front surface **404** and 4 mm away from the edge of back surface **406** where the length of top surface **412** and bottom surface **414** is 16 mm (e.g., the center line of the width is located at 8 mm from the edge of top surface **412** and bottom surface **414**). In some non-limiting embodiments, a first support block **402** may have through-hole **422** offset from the center line of the width of top surface **412** and bottom surface **414** and a second support block **402** may have through-hole **422** located along the center line of the width of top surface **412** and bottom surface **414**.

[0078] In some non-limiting embodiments, first color **424** may include white and/or a light color (e.g., off-white, yellow, tan, etc.). In some non-limiting embodiments, the first portion of support block **402** that includes first color **424** may be less than (e.g., less area, less volume, etc.) the second portion of support block **402** that includes second color **426**. In some non-limiting embodiments, the first portion of support block **402** that includes first color **424** may be made from polyoxymethylene (POM).

[0079] In some non-limiting embodiments, first color **424** may cover the surface area of cutout **418**. In some non-limiting embodiments, first color **424** may extend beyond the area of cutout **418** linearly forming an area that is rectangular in shape on top surface **412** adjacent to the surface area of cutout **418** and first color **424** may extend beyond the area of cutout **418** radially forming an area that is circular in shape on front surface **404** and back surface **406**. For example, first color **424** may extend 4.9 mm beyond the area of cutout **418** onto top surface **412** on each side of the area of cutout **418** forming a rectangular area of first color **424** on top surface **412**. First color **424** may extend in a 4.9 mm radial direction beyond the area of cutout **418** onto front surface **404** and back surface **406** forming a radial (e.g. circular, annular, etc.) area of first color **424** on front surface **404** and back surface **406**.

[0080] In some non-limiting embodiments, second color **426** may include black and/or a dark color (e.g., dark blue, dark brown, etc.). In some non-limiting embodiments, the second portion of support block **402** that includes second color **426** may be greater than (e.g., greater area, greater volume, etc.) the first portion of support block **402** that includes first color **424**. In some non-limiting embodiments, the second portion of support block **402** that includes second color **426** may be made from POM.

[0081] Referring now to FIG. 4B, FIG. 4B is a front view of a non-limiting embodiment of structure **400** for an inspection system. As shown in FIG. 4B, structure **400** may include at least two support blocks **402**. In some non-limiting embodiments, support block **402** may include front surface **404**, first side surface **408**, second side surface **410**, top surface **412**, bottom surface **414**, bored holes **416**, cutouts **418**, fastener holes **420**, through hole **422**, first color **424**, and second color **426**. In some non-limiting embodiments, a first portion of support block **402** may include first color **424** and a second portion of support block **402** may include second color **426**.

[0082] As further shown in FIG. 4B, each top surface **412** of each support block **402** is brought together such that top surfaces **412** of each support block **402** contact each other. As shown in FIG. 4B, when a first top surface **412** of each support block **402** is in contact with a second top surface **412**, cutouts **418** merge together to form an opening which may be configured to accept a tube. In some non-limiting embodiments, front view of structure **400** (e.g. one or more front surfaces **404**) may be a background in an image of a flange where the flange is superimposed on the background.

[0083] Referring now to FIG. 5, FIG. 5 is a diagram of a non-limiting embodiment of a medical

device **500**. In some non-limiting embodiments or aspects, medical device **500** may include tube **502**, elastomeric stopper **504**, distal end **506**, proximal end **508**, and flange **510**. In some non-limiting embodiments or aspects, tube **502** may include a barrel of a syringe, and/or the like. Tube **502** may be made of glass and/or plastic (e.g., glass and/or plastic suitable for medical applications). In some non-limiting embodiments or aspects, elastomeric stopper **504** may be made of an elastomeric material and may be adapted to be connected to medical device **500** (e.g., connected to a plunger rod of medical device **500**, and/or the like). In some non-limiting embodiments or aspects, distal end **506** may be adapted to be connected to another medical device (e.g., a needle, cap, catheter, stacked needle, and/or the like). In some non-limiting embodiments or aspects, proximal end **508** may include flange **510** and may be configured to receive elastomeric stopper **504**.

[0084] Referring now to FIG. **6A**, FIG. **6A** is a diagram of a non-limiting embodiment of an implementation **600** of inspection system **102**. As shown in FIG. **6A**, implementation **600** may include inspection system **102**, image capture device **104**, light source **106**, support blocks **402**, and tube **502**. In some non-limiting embodiments, support blocks **402** may include fastener holes **420**. In some non-limiting embodiments, fastener holes **420** may be used for mounting support blocks **402**.

[0085] In some non-limiting embodiments, support blocks **402** may be configured such that support blocks **402** may support tube **502** for inspection. In some non-limiting embodiments, tube **502** may include part of medical device **500** (e.g., a barrel of a syringe). Tube **502** may include at least one flange **510**. Inspection system **102** may inspect flange **510** of tube **502**. In some non-limiting embodiments, tube **502** may be positioned within support blocks **402**. In some non-limiting embodiments, front surfaces **404** of support blocks **402** may include first color **424** and second color **426**.

[0086] Light source **106** may be positioned such that light source **106** may illuminate flange **510** of tube **502**. In some non-limiting embodiments, light source **106** may be integrated with image capture device **104** and/or inspection system **102**. In some non-limiting embodiments, light source **106** may be separate from image capture device **104** and/or inspection system **102**. In some non-limiting embodiments, light source **106** may be positioned in between image capture device **104** and tube **502** such that light source **106** may illuminate flange **510** of tube **502** without obstructing a line of sight from image capture device **104** to tube **502**.

[0087] Referring now to FIG. **6B**, FIG. **6B** is a diagram of a non-limiting embodiment of another implementation **610** of inspection system **102**. As shown in FIG. **6B**, implementation **610** may include inspection system **102**, image capture device **104**, light source **106**, support blocks **402**, and tube **502**. In some non-limiting embodiments, inspection system **102** may include one or more of image capture device **104**, light source **106**, support blocks **402**, and/or tube **502**. In some non-limiting embodiments, implementation **610** may include a lens positioned between image capture device **104** and light source **106**.

[0088] In some non-limiting embodiments, support blocks **402** may be configured such that support blocks **402** may support tube **502** for inspection. In some non-limiting embodiments, tube **502** may include part of medical device **500** (e.g., a barrel of a syringe). Tube **502** may include at least one flange **510**. Inspection system **102** may inspect flange **510** of tube **502**. In some non-limiting embodiments, tube **502** may be positioned within support blocks **402**. In some non-limiting embodiments, front surfaces **404** of support blocks **402** may include first color **424** and second color **426**.

[0089] Light source **106** may be positioned such that light source **106** may illuminate flange **510** of tube **502**. In some non-limiting embodiments, light source **106** may be integrated with image capture device **104** and/or inspection system **102**. In some non-limiting embodiments, light source **106** may be separate from image capture device **104** and/or inspection system **102**. In some non-limiting embodiments, light source **106** may be positioned in between image capture device **104**

and tube **502** such that light source **106** may illuminate flange **510** of tube **502** without obstructing a line of sight from image capture device **104** to tube **502**. In some non-limiting embodiments, light source **106** may be positioned on the same axis as image capture device **104**. In some non-limiting embodiments, light source **106** may share a line of sight with image capture device **104** and may be positioned toward tube **502** in the same direction as image capture device **104**.

[0090] With regard to each of FIGS. 7A and 7B, as shown in FIGS. 7A and 7B, views of a non-limiting embodiment of captured images of an inspection system show an axial view (e.g., cross-sectional view) of a tube superimposed on different backgrounds in FIG. 7A, and a tube superimposed on a background with measurements of attributes of a flange of the tube being determined in FIG. 7B.

[0091] Referring now to FIG. 7A, shown is a view of a non-limiting embodiment of a comparison of images **700**. Comparison of images **700** includes first captured image **702** and second captured image **704**. First captured image **702** may include a background that is one color. As shown in FIG. 7A, first captured image **702** includes a tube superimposed on a fully white background. In this way, second captured image **704** may include a greater amount of contrast as compared to first captured image **702**, which allows for an inspection system (e.g., inspection system **102**) to more easily and accurately determine a measurement of an attribute of a tube, such as a diameter of a flange of the tube. In some non-limiting embodiments, second captured image **704** may represent an image that may be captured by inspection systems according to non-limiting embodiments of the present disclosure.

[0092] In some non-limiting embodiments, second captured image **704** may include a tube superimposed on a background. In some non-limiting embodiments, the background of second captured image **704** may include first color **708** and a second color **710**. In some non-limiting embodiments, first color **708** may include white and second color **710** may include black. As shown in second captured image **704** of FIG. 7A, first color **708** may surround the tube in second captured image **704** and second color **710** may surround first color **708**. As shown in second captured image **704** of FIG. 7A, first color **708** may form an annulus and/or circular area around the tube. It is to be understood that the shape formed by first color **708** in second captured image **704** may include any shape such that first color **708** surrounds the tube and such that second color **710** surrounds first color **708**.

[0093] As further shown in second captured image **704** of FIG. 7A, inspection system **102** may cause second captured image **704** to be captured. In some non-limiting embodiments, second captured image **704** may include first color **708** and second color **710**. In some non-limiting embodiments, second captured image **704** may display edge **706** of an internal diameter of a flange of the tube in second captured image **704**. In some non-limiting embodiments, edge **706** may appear clearer in second captured image **704** when compared to an edge of an internal diameter of a flange of the tube in first captured image **702**.

[0094] Referring now to FIG. 7B, FIG. 7B is a close-up view of a non-limiting embodiment of second captured image **704** of an inspection system (e.g., inspection system **102**) from comparison of images **700**. As shown in FIG. 7B, second captured image **704** may include first color **708** and edge **706**. In some non-limiting embodiments, inspection system **102** may perform an edge detection procedure on second captured image **704** of the flange of the tube to provide a defined edge of the flange of the tube. For example, inspection system **102** may perform an edge detection procedure on second captured image **704** by determining a measurement of an attribute of the flange of the tube such as the measurement of a diameter (e.g., an internal diameter or an external diameter) of the flange of the tube based on edge **706**.

[0095] Although embodiments or aspects have been described in detail for the purpose of illustration and description, it is to be understood that such detail is solely for that purpose and that embodiments or aspects are not limited to the disclosed embodiments or aspects, but, on the contrary, are intended to cover modifications and equivalent arrangements that are within the spirit

and scope of the appended claims. For example, it is to be understood that the present disclosure contemplates that, to the extent possible, one or more features of any embodiment or aspect can be combined with one or more features of any other embodiment or aspect. In fact, many of these features can be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of possible implementations includes each dependent claim in combination with every other claim in the claim set.

Claims

- 1.** A method for inspecting aspects of a medical device comprising: causing, with at least one processor, an image of a tube of a syringe to be captured via an image capture device, wherein the image comprises an image of a flange of the tube of the syringe superimposed on a background, wherein the background comprises a first color and a second color, wherein the first color of the background is white, and wherein the second color of the background is black; performing an edge detection procedure on the image of the flange of the tube; determining, with at least one processor, a measurement of a diameter of the flange of the tube based on the edge detection procedure performed on the image of the flange of the tube; and providing, with at least one processor, data associated with the measurement of the diameter of the flange of the tube.
- 2.** The method of claim 1, wherein performing the edge detection procedure on the image of the flange of the tube comprises: performing the edge detection procedure on the image of the flange of the tube to provide a defined edge of the flange of the tube; wherein determining the measurement of the diameter of the flange of the tube comprises: determining the measurement of the diameter of the flange of the tube based on the defined edge of the flange of the tube.
- 3.** The method of claim 1, further comprising: positioning the tube within a structure, wherein a surface of the structure comprises the first color and the second color.
- 4.** The method of claim 3, wherein the structure comprises a first component and a second component, and wherein the structure is configured to hold the tube between the first component and the second component.
- 5.** The method of any of claim 1, further comprising: causing light to illuminate the tube via at least one light source prior to causing the image of the tube to be captured via an image capture device, wherein the at least one light source comprises a spot light and a ring light.
- 6.** A system for inspecting aspects of a medical device, comprising: a structure configured to hold a tube for inspection; and at least one processor programmed or configured to: cause an image of a tube of a syringe to be captured via an image capture device when the tube is positioned in the structure, wherein the image comprises an image of a flange of the tube of the syringe superimposed on a background, wherein the background comprises a first color and a second color, wherein the first color of the background is white and wherein the second color of the background is black; performing an edge detection procedure on the image of the flange of the tube; determine a measurement of a diameter of the flange of the tube based on the edge detection procedure performed on the image of the flange of the tube; and provide data associated with the measurement of the diameter of the flange of the tube.
- 7.** The system of claim 6, further comprising: at least one light source, wherein the at least one light source is configured to illuminate the flange of the tube.
- 8.** The system of claim 6, wherein, when performing the edge detection procedure on the image of the flange of the tube, the at least one processor is further programmed or configured to: perform the edge detection procedure on the image of the flange of the tube to provide a defined edge of the flange of the tube; wherein, when determining the measurement of the diameter of the flange of the tube, the at least one processor is programmed or configured to: determine the measurement of the diameter of the flange of the tube based on the defined edge of the flange of the tube.

9. The system of claim 6, further comprising: a tube positioned within the structure, wherein a surface of the structure comprises the first color and the second color.

10. The system of claim 6, wherein the structure comprises a first component and a second component, and wherein the structure is configured to hold the tube between the first component and the second component.

11. The system of claim 7, wherein the at least one processor is further programmed or configured to: cause light to illuminate the tube via the at least one light source prior to causing the image of the tube to be captured via an image capture device, wherein the at least one light source comprises a spot light and a ring light.
